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REVIEW

Functional plasticity of pigment systems in microalgae and cyanobacteria under variable environmental conditions

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Abstract

Microalgae and cyanobacteria reconfigure pigment–protein complexes as environments change, coupling light harvesting, photochemistry, and redox homeostasis. This review synthesizes the mechanisms driving shifts in chlorophylls, carotenoids, and phycobiliproteins in response to abiotic and anthropogenic stressors, including irradiance, temperature, salinity, nutrient limitation, and pollutants. Pigment measurements are most informative when interpreted as profiles of individual pigments, absolute abundances, and diagnostically relevant ratios, and when analyzed together with photosystem performance and independent oxidative-damage markers. Key strategies include chlorophyll antenna remodeling to rebalance light capture and protection; carotenoid-mediated nonphotochemical quenching, excitation quenching, reactive oxygen species scavenging, membrane stabilization, and, in some taxa, secondary carotenoid accumulation during prolonged adaptation. In cyanobacteria, phycobiliprotein plasticity enables redistribution of excitation energy and supports antenna-level energy dissipation, while extracellular UV-screening pigments such as scytonemin provide an additional protective layer.

Keywords: experimental evolution; photoprotection; phototrophic microorganisms; pigment system; redox homeostasis.

Introduction

Photopigments of phototrophic microorganisms, including microalgae and cyanobacteria, are classically

described as structural and regulatory components of the photosynthetic apparatus that enable spectral photon absorption and transfer of excitation energy to reaction centers. This interpretation is sufficient to describe primary

Highlights

- Stress-driven pigment remodeling in microalgae and cyanobacteria
- Pigment profiles in combination with photosynthetic parameters for assessing photooxidative status
- Standardized assays and harmonized endpoints for pigment-based biomonitoring

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Abbreviations: ALE – adaptive laboratory evolution; APC – allophycocyanin; APX – ascorbate peroxidase; CAT – catalase; DNA – deoxyribonucleic acid; CTD – C-terminal domain; NTD – N-terminal domain; EPS – extracellular polymeric substances; FaRLiP – far-red light photoacclimation; F_v/F_m – maximum photochemical efficiency of PSII; FRL – far-red light; NADPH – nicotinamide adenine dinucleotide phosphate; LTEE – long-term evolution experiment; MDa – megadalton; NPQ – nonphotochemical quenching; OCP – orange carotenoid protein; PBPs – phycobiliproteins; PC – phycocyanin; PE – phycoerythrin; ROS – reactive oxygen species; SOD – superoxide dismutase; UV – ultraviolet; VDE – violaxanthin de-epoxidase.

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photochemical processes; however, it does not explain the persistent variability in pigment phenotypes observed in natural populations and in laboratory experiments under stress. Recent experimental and systems-level studies demonstrate that pigment systems function as a regulated component of cellular organization (Srivastava *et al.* 2022), embedded in redox-dependent regulatory circuits that coordinate environmental inputs with photosynthetic electron transport. In this framework, the pigment apparatus functions not only as an absorption module but also as a mechanism that maintains the operational stability of photosynthetic complexes under unstable environmental conditions.

In most cases, the link between stress and pigment-apparatus remodeling is mediated by control of excitation pressure and redox balance. When energy absorption exceeds the capacity for photochemical utilization and the available electron sinks, the probability of chlorophyll triplet-state formation and the generation of reactive oxygen species, including singlet oxygen, increases, leading to photooxidative damage of membranes and protein complexes. Therefore, quantitative and qualitative pigment changes should be interpreted as part of a coordinated regulatory response that redistributes excitation, limits reactive oxygen species (ROS) formation, and preserves photosystem stability under conditions in which redox homeostasis becomes vulnerable. In this context, photooxidative load, redox imbalance, and oxidative stress should be regarded as related but nonequivalent processes. Excess excitation energy increases photooxidative load, which may disturb cellular redox homeostasis and, when protective mechanisms become insufficient, culminate in oxidative stress and cellular damage (Shapiro *et al.* 2015, Allakhverdiev *et al.* 2024).

Regulation of pigment phenotypes is multilevel, reflecting coordinated changes in biosynthesis, complex assembly, and membrane architecture. Together, these processes generate the observed shifts in pigment composition and pigment ratios. At the biochemical level, changes in the balance between synthesis and degradation of tetrapyrroles and carotenoids determine the size of pigment pools. At the level of supramolecular organization, antenna size and the stoichiometry of pigment–protein complexes also vary, leading to changes in the distribution of excitation energy between light-harvesting complexes and reaction centers. In green algae and plants, light-harvesting complexes of photosystem II (LHCII) function as essential antenna systems involved in light harvesting, energy transfer, and photoprotection (Li *et al.* 2023). In diatoms, fucoxanthin chlorophyll (Chl) *a/c*-binding proteins (FCPs) perform these functions, whereas cyanobacteria possess large phycobilisome megacomplexes (6–16 MDa) (Zheng *et al.* 2021, Nagao *et al.* 2022). At the membrane level, thylakoid organization and antenna connectivity are remodeled, which affects energy-transfer kinetics and the local redox state in the vicinity of photosystems. Because these rearrangements proceed together with the induction of enzymatic and nonenzymatic antioxidant mechanisms, the pigment profile reflects an integral indicator of cellular physiological

state, rather than a proxy for biomass level or potential photosynthetic capacity (Nagarajan *et al.* 2022, Sun *et al.* 2024).

The physiological consequences of pigment rearrangements are also determined by cell-level optical factors. The effective absorption spectrum and the fraction of energy that is converted into photochemistry depend on pigment packaging, cell geometry, intracellular scattering, and self-shading. During acclimation and under stressors, these parameters change and shift the balance between photochemical energy use and its safe dissipation. Therefore, pigment sets that are similar in composition may yield different photobiochemical outcomes due to differences in optical path length, intracellular light gradients, and pigment–protein organization (Aizpuru and González-Sánchez 2024). This optical complexity is compounded in natural environments, where multiple simultaneous stressors alter both molecular pigment composition and cell-level optical properties through independent mechanisms.

Combined stresses provide the most informative context for analyzing pigment systems because, in natural environments, light regime, temperature, salinity, and pollution often vary jointly. Warming, salinization of inland waters, intensification of the ultraviolet component, and increasing metal pollution and xenobiotics create conditions under which the photosynthetic apparatus experiences simultaneous light-load pressure and constraints on electron transport (Allakhverdiev *et al.* 2024). In such situations, the increasing need for redox buffering is closely linked to the capacity of antioxidant and photoprotective systems to maintain cellular homeostasis (Shapiro *et al.* 2015). As a result, resilience is determined by how rapidly the cell increases the share of nonphotochemical dissipation, maintains photosystem repair, and stabilizes redox potential within a range compatible with operation of the electron transport chain (Sorathia *et al.* 2025). An illustrative example is the interaction of high irradiance and nitrogen starvation in *Haematococcus pluvialis* and *Dunaliella salina*. Under high light, lumen acidification activates the violaxanthin (V)–antheraxanthin (A)–zeaxanthin (Z) (VAZ) cycle, promoting the conversion of violaxanthin to zeaxanthin and contributing to rapid nonphotochemical quenching (NPQ) on a timescale of minutes to hours. Nitrogen starvation, acting over days, upregulates MEP-pathway genes and carotenogenic enzymes, including ketolases such as BKT1, thereby redirecting carbon flux toward secondary carotenoid biosynthesis and driving astaxanthin hyperaccumulation in *H. pluvialis* and β -carotene hyperaccumulation in *D. salina* (Zhao *et al.* 2020, Zarekarizi *et al.* 2023). Critically, the combination of both stressors elicits a stronger photoprotective and antioxidant response than either alone: elevated oxidative load under co-stress accelerates astaxanthin deposition, and analogous synergistic carotenoid enhancement under combined high light and nutrient deprivation has been independently confirmed in the chlorophyte *Coelastrella* sp., indicating that this two-phase, temporally layered response represents

a conserved strategy rather than a species-specific phenomenon (Liu *et al.* 2026).

The roles of individual classes of photopigments in this response differ, but they all serve the common purpose of coordinating light harvesting with photoprotection and antioxidant chemistry. Chlorophylls support primary photochemistry and at the same time define the risk of photodamage through the probability of triplet-state formation under excess excitation (Xue *et al.* 2020). Carotenoids combine light harvesting with chlorophyll-triplet and ROS quenching, while xanthophyll cycles provide rapid and reversible thermal dissipation under excess light (Jahns *et al.* 2009, Ramanna *et al.* 2017, Ma *et al.* 2019). Phycobiliproteins expand spectral coverage in cyanobacteria and some algae and enable plastic redistribution of excitation when the light regime changes and under chemical stress (Srivastava *et al.* 2022). Scytonemin belongs to specialized protective pigments and reduces ultraviolet radiation flux at the level of the cell envelope, supporting resilience under intense irradiation and under limited activity of energy-dependent photoprotective processes (Rastogi *et al.* 2014).

In practice, interpretation of pigment changes is complicated by the fact that responses with similar directionality can arise *via* different mechanisms and on different time horizons. Therefore, correct analysis requires distinguishing acclimation from adaptation. Acclimation refers to reversible changes in pigment parameters within the cell life cycle or short-term culture dynamics. Adaptation denotes stable intergenerational shifts that are fixed by selection and reproducible upon repeated exposures, manifested as stabilization of pigment phenotypes and regulatory regimes across generations (Hohmann-Marriott and Blankenship 2011, Patel *et al.*

2022). A complementary and equally critical interpretive axis runs between acclimatory and damage-mode responses. Both can produce externally similar pigment changes (e.g., declining chlorophyll, elevated carotenoid:chlorophyll ratios), but they differ fundamentally in the direction of cellular control. In the acclimatory mode, pigment changes are regulatory in origin: they are controlled by defined molecular circuits, proceed alongside maintained or improved photosystem performance (stable or recovering F_v/F_m), and are typically reversible when the stressor is removed. In the damage mode, photodamage accumulates faster than the capacity for repair and regulated dissipation; pigment changes are accompanied by declining F_v/F_m , rising lipid peroxidation markers such as malondialdehyde (MDA), and progressive loss of photosystem function. The transition between modes is determined by the balance between the rate of photodamage and the combined rate of photoprotection and PSII repair, and therefore depends on stress intensity, duration, and the combination of concurrent stressors (Muramatsu and Hihara 2012, Song *et al.* 2026). This distinction is particularly consequential under changing stress conditions: fluctuating irradiance, escalating pollutant doses, or simultaneous shifts in temperature and nutrients, because such dynamics can drive the system across the acclimation-damage threshold repeatedly and in both directions on timescales of minutes to days. Throughout this review, distinguishing these modes and identifying the multiparameter combinations of pigment, photobiophysical, and oxidative-damage indicators that operationalize the distinction is a unifying analytical objective.

Fig. 1 shows that diverse environmental stressors converge on a common functional challenge, coordinating light capture with photooxidative risk through the balance

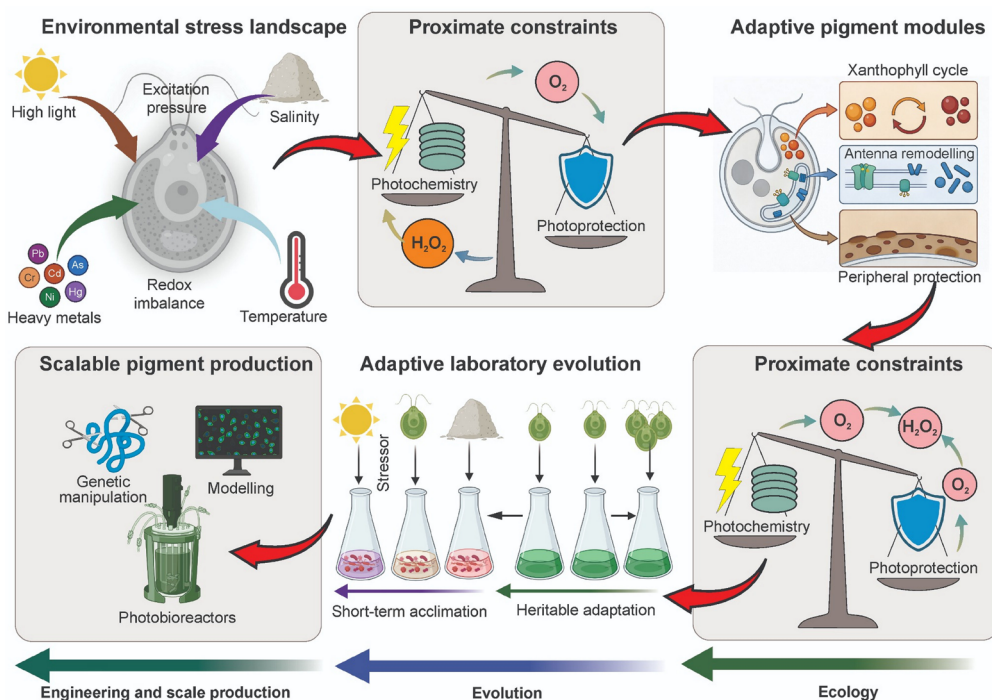


Fig. 1. Conceptual scheme of stress-induced remodeling of phototroph pigment systems, from the stressor landscape to adaptive modules and diagnostic readouts.

between photochemistry and photoprotection. This task can be solved in a modular manner, combining retuning the carotenoid module, changes in effective antenna size, and screening pigments, which enables functional convergence despite multiple molecular implementations. In applied contexts, the same pigment and redox-linked traits provide a basis for biomonitoring and for diagnosing stress load in natural and engineered systems.

This narrative review examines systematically how photopigment systems in microalgae and cyanobacteria are remodeled under different stressors. Rather than providing a purely descriptive overview, we present a mechanistic synthesis of how pigment systems regulate excitation pressure, photoprotection, and redox balance. We examine how pigment modules shape energy partitioning, reactive oxygen species formation, and photosystem stability, and why similar pigment responses may reflect different physiological scenarios across taxa and conditions. The review is organized into functional blocks, including chlorophylls, carotenoids, phycobiliproteins, and extracellular UV screens, and integrates evolutionary perspectives and experimental-evolution evidence. We conclude by outlining key remodeling patterns, current limitations, and priorities for improving comparability and causal interpretation.

Chlorophyll system and regulation of light harvesting under photooxidative load

Chlorophylls form the core of the photochemical apparatus in microalgae and define the link between light-field characteristics and the metabolic output of photosynthesis. As constituents of light-harvesting complexes and reaction centers, they enable transfer of excitation energy to the primary charge-separation reactions (Hill *et al.* 2005). At the same time, the chlorophyll system is coupled to the risk of photodamage. Under excess absorbed energy, the probability of intersystem crossing and the formation of reactive oxygen species, primarily singlet oxygen, increases, thereby making chlorophyll a potential source of photooxidative load (Ruban 2016). Therefore, photosynthetic resilience is determined not by maximizing the chlorophyll pool, but by fine-tuning its amount and organization toward a balance between light harvesting and photoprotection, which depends on the structural and functional state of the antenna system.

Observed ranges of chlorophyll content confirm the high plasticity of this system. Under favorable conditions, the total chlorophyll fraction often falls within 0.5–1.0% of dry biomass, although substantially higher values have been reported for some fast-growing chlorophytes under optimized cultivation; for *Chlorella vulgaris*, chlorophyll accumulation up to 7% of dry mass has been reported (D'Alessandro and Antoniosi Filho 2016). These parameters are sensitive to light intensity and spectral composition, temperature, pH, salinity, nutrient availability, and the presence of xenobiotics and heavy metals. For this reason, chlorophyll is widely used as a quantitative indicator of physiological state. As a component of aquatic environmental quality

assessment, its interpretation requires accounting for the stress regime and the culture growth stage (Koutra *et al.* 2018).

Under stress, chlorophyll dynamics do not follow a single direction because they reflect the relationship between growth limitation and redistribution of resources. For example, in *Scenedesmus obliquus*, exposure to the dye *Brilliant Green* was accompanied by partial growth inhibition and pronounced changes in chlorophyll content together with increased protein and lipid fractions, which indicates stress-induced redistribution of carbon and energy fluxes (Rajamanickam and Selvasembian 2026). In other cases, sublethal xenobiotic concentrations may temporarily increase chlorophyll content and biomass. For *Chlorella* sp., exposure to imidacloprid at 10–50 mg L⁻¹ was associated with increased chlorophyll content and dry mass after 12 d of cultivation, which is consistent with a hormetic scenario until a toxicity threshold is reached (Deng *et al.* 2022). Therefore, changes in the chlorophyll pool should be evaluated together with markers of oxidative stress and indicators of metabolic reallocation, because an externally similar trend may correspond either to damage or to compensatory remodeling. At the gene-regulatory level, chlorophyll biosynthesis is gated at the Mg-chelatase step, where the catalytic subunit CHLH (GUN5 in eukaryotes) inserts Mg²⁺ into protoporphyrin IX to commit metabolic flux irreversibly to the chlorophyll branch; loss-of-function *gun5/cch* mutations abolish Mg-chelatase activity and drastically reduce chlorophyll accumulation, confirming CHLH as the principal regulatory node of the pathway (Beale 2009). Activity of this step is enhanced by GUN4, a porphyrin-binding protein that associates with both protoporphyrin IX and Mg-protoporphyrin IX, stimulates Mg-chelatase *in vitro* and *in vivo* by promoting CHLH membrane association, and facilitates substrate-to-product transfer through the Mg-branch, thereby coupling enzymatic activation to metabolic flux control. Additionally, GUN4–porphyrin complexes generate ¹O₂ under illumination, positioning GUN4 at the interface between chlorophyll biosynthesis and retrograde signaling (Adhikari *et al.* 2011). Under excess light, both H₂O₂ and ¹O₂ function as plastid-to-nucleus retrograde messengers: H₂O₂ exits the chloroplast to modulate nuclear gene expression during high-light stress, while ¹O₂, acting through the EX1/EX2-mediated pathway, broadly represses chloroplast-function and tetrapyrrole-biosynthesis genes, with CHLH/GUN5 itself participating in this signaling network through physical interaction with EX1/EX2, such that *gun5* mutations suppress ¹O₂-triggered retrograde responses and alter the downregulation of nuclear photosynthetic genes, although direct experimental evidence that *chlH* transcript levels are preferentially repressed by H₂O₂ or ¹O₂ has not yet been explicitly documented, the available evidence positions retrograde ROS signaling as a mechanistically coherent suppressor of *chlH* transcription independently of growth rate (Leister 2019). This retrograde regulatory logic extends to the transcriptional level, where sigma factor identity determines which subset of photosynthetic and chlorophyll-related genes is activated in response to

environmental context: in cyanobacteria, alternative σ subunits redirect RNA polymerase to distinct promoter sets in response to light shifts, nutrient status, and oxidative stress, with group 2 sigma factors collectively modulating stress survival, nutrient acquisition, and photosynthetic competence (Srivastava *et al.* 2020). In *Synechococcus elongatus* PCC 7942, light-upshift induces *rpoD3* (SigD) expression alongside high-light-responsive loci including *hliA* and *nblA* (Yasuda *et al.* 2020), and in *Synechocystis* sp. PCC 6803, blue light drives coordinated transcriptional remodeling of PSII core subunits (*psbA*, *psbD*, and *psbC*), identifying SigD as an integral component of the light-responsive module that aligns antenna and reaction-centre gene expression with irradiance (Luimstra *et al.* 2020). Nitrogen status operates through a parallel axis, remodeling chlorophyll-binding protein stoichiometry and phycobilisome composition through environmentally responsive regulators – among them, putatively, SigE (*rpoD4*) – thereby coupling nitrogen sufficiency to chlorophyll biosynthesis and antenna assembly, though direct evidence for SigE or SigD occupancy at chlorophyll biosynthesis promoters has yet to be reported.

Plasticity of the chlorophyll system is determined not only by quantity, but also by chlorophyll type, which expands the spectral capabilities of photosynthesis and affects excitation partitioning. In photosynthetic eukaryotes and cyanobacteria, chlorophylls *a*, *b*, *c*, *d*, and *f* have been described, differing in spectral properties and taxonomic distribution (Gisriel 2024, Tamiaki and Kichishima 2025). Chlorophyll *a* is universal and serves as the primary photochemical pigment. Chlorophyll *b* is characterized by a formyl group at the C-7 position and is among the most abundant chlorophylls in oxygenic phototrophs after chlorophyll *a*. Both pigments are embedded in photosystems, large supramolecular protein–pigment complexes that support primary photochemical

reactions (Kume *et al.* 2018). Chlorophyll *c*, typical of diatoms, corresponds to aquatic optical conditions with a shift of the available spectrum toward the blue region (Büchel 2020). The most specialized forms, chlorophylls *d* and *f*, have been identified in a limited number of cyanobacteria and enable use of the far-red range above 700 nm, which increases photosynthetic performance in spectrally filtered habitats where the visible component of light is attenuated (Elias *et al.* 2024). Thus, the taxonomic mosaic of chlorophylls reflects evolutionarily fixed solutions that broaden the spectral window of photosynthesis and reduce the risk of excitation imbalance by redistributing absorption across wavelengths. Table 1 summarizes spectral features of chlorophylls in microalgae and cyanobacteria, their functional role in light harvesting, and typical directions of change in the content and ratios of these pigments under varying environmental conditions (Hill *et al.* 2005, Ruban 2016, Koutra *et al.* 2018, Croce and van Amerongen 2020, Tamiaki and Kichishima 2025).

Photosystem pigment composition and antenna organization differ fundamentally between cyanobacteria and eukaryotic microalgae, and these architectural differences determine how each group responds to environmental stress at the level of individual photosystems. In cyanobacteria, light harvesting relies on phycobilisomes – extrinsic antenna complexes assembled from an allophycocyanin core and variable phycocyanin/phycoerythrin rods that are physically coupled to the cytoplasmic face of the thylakoid membrane and transfer excitation to both PSII and PSI. Because phycobilisomes are soluble and mobile, their composition is remodeled rapidly in response to stress: chromatic acclimation switches rod pigmentation between phycocyanin and phycoerythrin in response to light quality, salt, pH, sulfur limitation, and pollutant exposure drive shifts in chlorophyll *a* content and PC/PE stoichiometry, and under pollutant-induced ROS

Table 1. Spectral and functional types of chlorophylls in microalgae and cyanobacteria.

Chlorophylls	Dominant color, absorption maximum	Present in species or groups	Main function in microalgae	Response to stress factors
Chlorophyll <i>a</i>	Blue-green, 660–665 nm	Microalgae and cyanobacteria, for example <i>Chlorella</i> sp., <i>Spirulina</i> sp., <i>Scenedesmus</i> sp., <i>Microcystis aeruginosa</i> , <i>Pavlova lutheri</i>	Core photochemical pigment of oxygenic photosynthesis, supports energy capture and excitation transfer to reaction centers	Content varies depending on light, nutritional status, and growth phase, often used as an indicator of physiological state
Chlorophyll <i>b</i>	Green or yellow, 642–652 nm	Green algae Chlorophyta and euglenoids Euglenophyta	Accessory pigment, expands the absorption spectrum and transfers energy to chlorophyll <i>a</i>	Content is species-specific and varies under environmental factors
Chlorophyll <i>c</i>	Yellow green, approximately 630 nm	Diatoms and some chromists	Light-harvesting pigment adapted to the spectral conditions of aquatic environments	Determined by species, nutritional status, and environmental conditions
Chlorophyll <i>d</i>	Bright green, approximately 696 nm	Selected cyanobacterial lineages and phototrophs associated with far-red light environments	Expands the use of far-red light	Associated with adaptation to spectrally filtered niches
Chlorophyll <i>f</i>	Emerald green, approximately 707 nm	Cyanobacteria	Supports photosynthesis in the far-red range	Inducible adaptation to far-red light

accumulation, phycobilisome integrity is compromised at the allophycocyanin core, impairing excitation transfer to PSII and reducing the number of active PSII centers (Zhang *et al.* 2024). Iron limitation represents a particularly instructive case: it depresses chlorophyll-rich PsaA/PsaB subunit levels, directly reducing PSI-associated pigment content and compromising PSI electron flow, while PSII genes may remain comparatively stable, revealing that PSI is not simply a passive downstream recipient of antenna-level damage but an independent iron-sensitive target. In eukaryotic microalgae, antenna function is performed by membrane-integral LHC complexes (LHCII and LHCI in the green lineage) built from chlorophyll *a/b*-protein assemblies, and stress-driven reorganization of these complexes is reflected in shifts of the Chl *a/b* ratio, which serves as a quantitative index of antenna resizing: increasing Chl *a/b* under oxidative or high-light stress indicates selective loss of Chl *b*-enriched peripheral LHCII, reducing effective antenna cross-section (Singh *et al.* 2026). High carotenoid accumulation accompanies this process, increasing the carotenoid-to-chlorophyll ratio and providing both photoprotective quenching capacity and reduced PSII antenna size; under extreme irradiance in some *Chlorella* species, the PSII–LHCII superstructure itself is reconfigured, indicating that antenna remodeling extends beyond pigment stoichiometry to the physical organization of the photosystem supercomplexes (Scarponi *et al.* 2025, Steiner *et al.* 2026).

PSII is the primary site of photodamage under abiotic stress in both groups, but the mechanisms and downstream consequences differ in instructive ways. In cyanobacteria, high light, ROS, chemical pollutants, and acidification converge on PSII by disrupting D1 protein integrity and impairing excitation delivery from the phycobilisome core, thereby causing a rapid decline in PSII activity that precedes broader metabolic collapse. This reduction in antenna input occurs in a defined sequence, as phycobilisome disassembly is characterized by the preferential release of rod pigments before degradation of the allophycocyanin core, with this process being mediated by the NblA proteolytic adaptor (Tokodi *et al.* 2025). PSI, by contrast, shows greater stability under most stress conditions and may even maintain or upregulate associated gene expression when PSII is impaired, though iron limitation overrides this pattern by specifically targeting the chlorophyll-rich PSI reaction centre. State transitions provide a further regulatory dimension unique to cyanobacteria: phycobilisome migration between PSII and PSI redistributes excitation energy on timescales of seconds to minutes, allowing the cell to balance photosystem loading before transcriptional or proteolytic remodeling is engaged (Steiner *et al.* 2026). In eukaryotic microalgae, PSII–LHCII connectivity is regulated through LHCII phosphorylation, a process triggered by plastoquinol accumulation under excess light or elevated CO₂. Phosphorylation promotes dissociation of LHCII from PSII, thereby reducing PSII quantum yield and redistributing antenna excitation toward PSI. This mechanism functionally resembles a state transition, although it is mediated through covalent

modification of antenna proteins rather than through the physical migration of a soluble light-harvesting complex (Scarponi *et al.* 2025). Chemical stress and metal toxicity in microalgae produce PSII damage accompanied by chlorophyll loss, paralleling the cyanobacterial pattern, but without the ordered phycobilisome disassembly sequence; instead, loss of Chl *b* and peripheral LHCII components precedes core antenna degradation, reflecting the structural logic of the LHC system. Across both groups, the consistent pattern is that PSII antenna capacity is reduced before reaction-centre integrity is lost, whether through carotenoid-dependent quenching, antenna protein remodeling, or phycobilisome proteolysis – a convergent response that limits singlet-oxygen production at the cost of reduced photochemical throughput (Sharma *et al.* 2025).

Beyond changes in chlorophyll content, some phototrophs exhibit qualitative remodeling of chlorophyll composition, enabling spectral tuning of light absorption and maintenance of photosynthetic function under spectrally constrained environments. Chlorophyll *f* is the most illustrative example: it extends the operational range of oxygenic photosynthesis into the far-red region above 700 nm and makes available photons that are poorly usable for most oxygenic phototrophs (Tros *et al.* 2021). It was described in cyanobacteria associated with Shark Bay stromatolites and is considered a rare biochemical innovation that is functionally relevant under pronounced spectral filtering. The producer of chlorophyll *f* was identified as *Halomicronema hongdechloris* (Chen *et al.* 2012). The spectral shift is linked to a formyl group at the C-2 position and allows chlorophyll *f* to be classified as 2-formyl-chlorophyll *a*, which is accompanied by pronounced photobiophysical consequences with minimal chemical modification of the chlorophyll scaffold.

Chlorophyll *f* function is associated with the far-red light acclimation program Far-Red Light Photoacclimation (FaRLiP), in which approximately 8–12.5% of cellular chlorophyll *a* is reversibly replaced by chlorophyll *f*. The remodeling affects not only antenna regions, but also components of both photosystems, including the primary electron donors. Importantly, FaRLiP is an inducible mode and differs from constitutive far-red photosynthesis based on chlorophyll *d* in *Acaryochloris* (Patel *et al.* 2025). As a result, a mixed system is formed with dominance of chlorophyll *a* and a functionally critical contribution of chlorophyll *f* despite its low abundance, sometimes with the presence of trace amounts of chlorophyll *d* (Antonaru *et al.* 2025). This principle emphasizes that spectral adaptation is implemented through remodeling of photosynthetic apparatus architecture rather than through simple accumulation of a single pigment. The ecological logic of FaRLiP matches conditions in which the visible part of the spectrum is attenuated due to shading, competition among phototrophs, or light passage through absorbing matrices. Such systems include dense microbial mats and stromatolites, shaded rock surfaces, soil crusts, and some thermal habitats, where the far-red range becomes relatively enriched (Murray *et al.* 2022). At the molecular level, FaRLiP relies on a specialized genetic cluster that includes approximately 15–21 genes

and encodes paralogs of PSI, PSII, and phycobilisome components. This supports coordinated remodeling of photosynthetic complexes (Gan *et al.* 2014). In ecological and metagenomic studies, identification of far-red light-adapted cyanobacteria most often uses detection of the gene *chlF* encoding chlorophyll *f* synthase (Di Stefano *et al.* 2024). Proteomic data show that upon a shift to far-red light, the primary remodeling occurs in photosynthetic complexes, whereas central metabolic pathways remain relatively stable, which is consistent with a narrowly directed program of spectral acclimation (Chen *et al.* 2019). Transcription of the FaRLiP cluster is controlled by the dedicated far-red-sensing phosphorelay system RfpA/RfpB/RfpC. The phytochrome-class photoreceptor RfpA detects far-red light and initiates signal transduction through autophosphorylation, followed by phosphotransfer to the shuttle protein RfpC and subsequently to the response regulator RfpB. Once phosphorylated, RfpB activates transcription of FaRLiP-associated genes, thereby establishing one of the best-characterized light-quality-responsive regulatory programmes described in cyanobacteria (Ho *et al.* 2017).

An important implication is the diagnostic value of chlorophyll *f* as a marker of far-red light adaptation. Its synthesis is induced by far-red wavelengths and reverses upon return to white light, enabling detection of FaRLiP activity in cultures and environmental samples (MacGregor-Chatwin *et al.* 2022). Taken together, the strongest evidence indicates that chlorophyll-based acclimation depends primarily on antenna organization, pigment distribution between antennae and reaction centers, and PSI–PSII stoichiometry rather than on total chlorophyll content alone. Consequently, changes in bulk chlorophyll cannot independently distinguish regulated acclimation from pigment degradation and photodamage. Although antenna resizing and FaRLiP are well-established mechanisms, direct causal links between ROS signaling and the transcriptional control of individual chlorophyll-biosynthesis genes remain less clearly demonstrated across taxa. Future studies should therefore combine pigment composition with antenna cross-section, PSI/PSII performance, repair kinetics, and oxidative-damage markers.

Carotenoid system of photoprotection and redox buffering under stress factors

Carotenoids represent one of the most functionally versatile pigment classes in microalgae and cyanobacteria. Their role is not limited to auxiliary light harvesting, because carotenoids account for a major fraction of the molecular protection of the photosynthetic apparatus against the consequences of excess excitation (García-Oneto *et al.* 2024, Mostafa and Hashem 2025). By chemical criteria, carotenoids are typically subdivided into carotenes, including α -carotene and β -carotene, and xanthophylls, including lutein and violaxanthin, reflecting the presence of oxygen-containing groups and differences in membrane distribution and functional contributions to photoprotection and redox balance (Kolašinac *et al.* 2021).

Within light-harvesting complexes, carotenoids participate in energy transfer, and as excitation pressure increases, they support dissipation of excess energy, quenching of chlorophyll triplet states, and neutralization of radical species *via* electron transfer, radical adduct formation, and hydrogen atom transfer. For correct interpretation of stress responses, it is useful to distinguish carotenoids that are constitutively associated with photosynthetic complexes from accumulation-type carotenoids that are stress-induced. Primary carotenoids are integral to antennae and the microenvironment of reaction centers and directly contribute to matching light harvesting to the capacity for photochemical energy use. Secondary carotenoids accumulate under prolonged adverse exposure and function as optical and redox reserves, often being deposited in specialized intracellular structures (Shi *et al.* 2020).

In green algae, experimental reduction of antenna size improved light-use efficiency and biomass productivity, indicating that pigment organization directly affects the distribution of absorbed energy within the photosynthetic apparatus (Cazzaniga *et al.* 2014, Negi *et al.* 2020). Under excess light, photoprotective dissipation also helps limit PSI photoinhibition and the associated risk of reactive oxygen species formation (Takagi *et al.* 2019). In cyanobacteria, carotenoid-triggered quenching in phycobilisomes has been shown experimentally to divert excitation away from both photosystems (Rakhimberdieva *et al.* 2010). In this sense, pigment-dependent energy dissipation should be viewed not only as photoprotection, but also as part of the regulation of electron flow. This distinction helps explain why the total carotenoid pool does not show a universal increase under higher irradiance. In a study on *Pseudoneochloris marina*, carotenoid content decreased as light intensity increased (Gonçalves *et al.* 2019). Such results indicate that response direction is determined not only by light load, but also by exposure duration, by which carotenoid subclasses are included in the total estimate, and by how antenna complex composition changes during acclimation.

Notably, carotenoid regulation is often branch-specific and reflects functional partitioning between light harvesting and photoprotection. In the euryhaline microalga *Tetraselmis* sp. CTP4, different branches of the carotenoid pathway respond in opposite directions to changes in irradiance. β -carotene content was up to threefold higher under low light at $33 \mu\text{mol}(\text{photon}) \text{m}^{-2} \text{s}^{-1}$ than under higher irradiance, which is consistent with its contribution to maintaining efficient energy capture under photon limitation. In contrast, lutein increased by approximately 1.5-fold under high light at $170\text{--}280 \mu\text{mol}(\text{photon}) \text{m}^{-2} \text{s}^{-1}$, which corresponds to strengthened nonphotochemical protection. At the same time, neoxanthin and violaxanthin showed levels that were 2–3-fold higher under low light, indicating that even within the xanthophyll pool, regulation can reflect distinct functional demands depending on the photobiophysical regime. These data emphasize that interpreting total carotenoids as a single indicator of stress tolerance is often insufficient without considering the specific carotenoid profile and the conditions under

which it is formed. The fastest-acting level of carotenoid photoprotection is associated with xanthophyll cycles, which set the potential for nonphotochemical quenching (NPQ) and limit photoinhibition. In higher plants, the xanthophyll cycle is triggered by lumen acidification and, *via* violaxanthin de-epoxidase (VDE), converts violaxanthin to zeaxanthin through the VAZ cycle (Ruban 2016). The cycle is initiated when light-driven proton pumping acidifies the thylakoid lumen to pH 6.2, at which point VDE undergoes protonation-dependent conformational opening, dimerizes, and associates with MGDG-rich domains of the luminal membrane surface to gain access to its lipophilic substrates; the enzyme then catalyzes stepwise de-epoxidation utilizing ascorbate as the obligate electron donor (Jahns *et al.* 2009, Fufezan *et al.* 2012). The zeaxanthin produced is incorporated into LHC complexes, where it modifies pigment–protein interactions to activate NPQ pathways that dissipate excess excitation energy as heat, while simultaneously protecting thylakoid membrane lipids from peroxidation. Upon a decrease in irradiance and the consequent rise in luminal pH, stromal ZEP catalyzes the back-reaction [zeaxanthin (Z)-antheraxanthin (A)-violaxanthin (V)] using NADPH and O₂ as co-substrates. The intrinsically slower kinetics of ZEP relative to VDE ensure a controlled relaxation of the NPQ state, preventing premature restoration of full antenna absorption capacity before irradiance has sufficiently declined, which collectively enhances NPQ and reduces reactive oxygen species formation (Berne *et al.* 2018, Zhou *et al.* 2020). In microalgae, the contribution of this cycle is taxon-specific. In diatoms, NPQ is linked to the diadinoxanthin (DD) diatoxanthin (Dt) cycle, in which DD is de-epoxidized to Dt by a VDE-type diadinoxanthin de-epoxidase that can operate at higher luminal pH than canonical VDE, including in model species such as *Phaeodactylum tricornutum* and *Thalassiosira pseudonana*, whereas in chlorophytes alternative de-epoxidases may occur and zeaxanthin can have a more pronounced antioxidant role (Jakob *et al.* 2001, Girolomoni *et al.* 2020). In both cases, shortening the lifetime of excited states and reducing effective absorption of PSII decrease the probability of singlet oxygen formation and thereby limit the development of photooxidative cascades (Giossi *et al.* 2020). When comparing studies, it is important to consider that NPQ measurements alone do not resolve the mechanism if data on the sizes of relevant xanthophyll pools and interconversion kinetics are lacking, because the limiting factor may be pigment availability, enzymatic capacity, or antenna architecture.

Carotenoids also perform structural and protective functions that are particularly relevant under salt stress, temperature extremes, and nutrient limitation. Located in thylakoid membranes and pigment–protein complexes, carotenoids influence the physical properties of the membrane phase and help maintain supercomplex integrity. This effect is strengthened when lipid composition is remodeled and membrane viscosity changes in a coordinated manner (Joshi *et al.* 2023, Song *et al.* 2024). Under nutrient limitation, when degradation of thylakoid lipids and proteins increases photosystem

vulnerability, carotenoid-dependent excitation quenching and ROS detoxification can slow the development of photoinhibition and support functional membrane organization (Muzzopappa and Kirilovsky 2020). Thus, the carotenoid profile reflects an indicator linked not only to photochemical parameters, but also to the stability of membrane architecture.

A distinct form of carotenoid adaptation is stress-induced hyperaccumulation of secondary carotenoids, well documented for astaxanthin in *Haematococcus pluvialis* and β -carotene in *Dunaliella salina*. In both systems, high light and nutrient deficiency activate carotenogenesis coupled to triacylglycerol accumulation, with lipid droplets serving as the primary depot for carotenoid sequestration (Aditi *et al.* 2025). In *H. pluvialis*, this response is driven by transcriptional upregulation of MEP-pathway genes, including *DXS*, *CMS*, *CMK*, *HDS*, and *HDR*, which supply isoprenoid precursors for the sequential oxidative conversion of β -carotene to astaxanthin: β -carotene ketolase (BKT, encoded by stress-induced paralogs *bkt1*, *bkt2*, and *bkt3*, upregulated up to 52-fold) catalyzes ketolation through echinenone to canthaxanthin, while β -carotene hydroxylase (CrtR-b, with the additionally characterized CrtR-b2 isoform) drives parallel hydroxylation through β -cryptoxanthin and zeaxanthin, with convergent intermediates yielding adonixanthin and finally astaxanthin (Liang *et al.* 2016). In *D. salina*, β -carotene accumulates within chloroplast lipid globules (plastoglobuli) at concentrations reaching up to 10% of dry mass under high-light and nutrient stress, with expansion of this storage compartment tightly correlated with *de novo* pathway upregulation and fatty acid remodeling; plastoquinone-derived isoprenoid antioxidants, including plastochromanol-8, contribute to redox buffering and oxidative stress protection in this membrane environment, though their role as direct coordinators of β -carotene partitioning to plastoglobuli has not been experimentally established (Kim *et al.* 2024). The practice of two-stage cultivation that includes biomass growth under favorable conditions followed by controlled stress induction reflects a trade-off between cell division and redirection of metabolic fluxes into protective reserves, which increases final pigment productivity (Gao *et al.* 2017). To compare responses across taxa, Table 2 provides examples of carotenoid shifts under different stressors, indicating response direction and functional interpretation at the level of photoprotection and redox buffering.

Carotenoids are involved not only in effector, but also in regulatory layers of the stress response. They are linked to signaling cascades, including *via* hormone-like derivatives and *via* redox-dependent tuning of carotenogenesis (Sun *et al.* 2022). In *Dunaliella salina*, ROS associated with the activity of NADPH oxidase homologs were linked to induction of β -carotene biosynthesis under high irradiance, which supports the view of ROS as regulatory signals that set the scale of pigment investment when conditions change (Kim *et al.* 2024). As a result, the carotenoid phenotype is shaped as an outcome of feedback between excitation pressure, redox state, and metabolic fluxes, rather than as an isolated marker of exposure.

Table 2. Carotenoid responses of microalgae under stress exposures.

Strain	Stressor	Carotenoid response	Functional interpretation in the stress response	Reference
<i>Mychonastes</i> sp.	High antibiotic concentrations (sulfamethoxazole)	Increase in total carotenoid content	Strengthening of nonenzymatic antioxidant protection	Eheneden <i>et al.</i> (2024)
<i>Chlorella</i> sp. L166, <i>Scenedesmus quadricauda</i>	Antibiotic (cefradine)	Increase in carotenoids	Limitation of lipid peroxidation and increased cellular resilience	Wang <i>et al.</i> (2022)
<i>Chlorella thermophila</i> MF179624	High antibiotic concentrations (azithromycin)	Gradual increase in carotenoid content	Quenching of excited states and limitation of secondary lipid peroxidation products	Mohanty and Mohanty (2025)
<i>Dunaliella salina</i>	High light, salinity, N limitation	Increase in β -carotene (up to 13% DCW)	Optical screening and antioxidant buffering under high ROS load	Keramati <i>et al.</i> (2021)
<i>Coelastrella</i> sp.	Magnetic field	Marked decrease in carotenoid content	Reduction of the nonenzymatic antioxidant pool and potentially weakened ROS buffering	López-Bello <i>et al.</i> (2025)
<i>Mychonastes</i> sp. 247	Light intensity, salinity	Increased production of lutein and zeaxanthin	Enhanced energy dissipation and photoprotection of the photosynthetic apparatus	Hong <i>et al.</i> (2023)
<i>Chlorococcum humicola</i>	CO ₂ (1–3% v/v), Lux blue light	Marked increase in β -carotene and astaxanthin	Enhanced carbon fixation and increased availability of carotenoid biosynthesis precursors	Kunyawut <i>et al.</i> (2025)
<i>Muriellopsis</i> sp.	Ecological stress	Increase in lutein	Strengthening of the antioxidant response under oxidative pressure	Cruz-Balladares <i>et al.</i> (2021)
<i>Coelastrella saipanensis</i>	Milk and fish wastewater	Increase in β -carotene and astaxanthin	Increased antioxidant activity and resilience to combined exposure	Karicheri and Kulanthaiyesu (2025)

At the transcriptional level, oxidative signals reach carotenogenic promoters through a 2-Cys peroxiredoxin (2-Cys Prx)-mediated redox relay: 2-Cys Prx reacts rapidly with H₂O₂, undergoes an oxidation-dependent conformational change, and transfers oxidative equivalents to thioredoxin (Trx), thereby modulating the redox state of Trx-dependent transcriptional regulators (Winterbourn 2025). This cascade is consistent with ROS-dependent upregulation of *psy* (phytoene synthase), *pds* (phytoene desaturase), and *bkt* (β -carotene ketolase) observed in *Haematococcus pluvialis* under high light, nsPEF treatment, and chemical oxidative inducers, although direct experimental evidence that oxidized 2-Cys Prx interacts with or modulates transcription specifically from *psy*, *pds*, or *bkt* promoters in this organism has not yet been reported (Mallén-Ponce *et al.* 2024). A complementary post-translational layer is provided by the ORANGE (Or) protein, a plastid-localized chaperone that stabilizes phytoene synthase independently of transcript abundance, ensuring that carotenoid flux increases before full transcriptional upregulation is established (Kumari *et al.* 2020).

Overall, the most firmly established carotenoid functions are xanthophyll-cycle-dependent energy dissipation, chlorophyll-triplet quenching, and protection against oxidative damage. By contrast, total carotenoid content is often an ambiguous stress indicator because individual carotenoid branches may respond in opposite directions and on different timescales. Stress-induced hyperaccumulation of astaxanthin or β -carotene

represents prolonged metabolic reallocation rather than the rapid photoprotective response mediated by primary xanthophylls. Regulatory connections involving ROS, 2-Cys peroxiredoxin, and the ORANGE protein remain promising but are not equally supported by direct evidence in all microalgal taxa. Pigment-resolved and time-resolved analyses are therefore required for mechanistic interpretation.

Phycobiliproteins and plasticity of the light-harvesting antenna under stress conditions

Phycobiliproteins (PBPs) represent a specialized class of photopigments that largely determine acclimation and resilience strategies in cyanobacteria and in some algal lineages. In contrast to membrane-bound chlorophylls and carotenoids, PBPs are water-soluble chromoproteins that contain open-chain tetrapyrrole chromophores (phycobilins), a structural organization that defines their spectral properties and functional role in light harvesting (Dumay and Moránçais 2016). This molecular organization also confers high structural flexibility to the phycobiliprotein antenna and enables rapid adjustments of its quantitative parameters and functional properties in response to changes in irradiance, spectral regime, and chemical stressors (Stadnichuk 2025).

Key PBPs include phycocyanin (PC), allophycocyanin (APC), and phycoerythrin (PE). These pigments broaden the spectral window of photosynthesis because they efficiently absorb radiation in the green-to-orange

region at 450–650 nm, which is used less efficiently by chlorophyll *a* (Jha *et al.* 2024). However, the functional significance of PBPs in stress physiology is not limited to spectral complementarity. Experimental studies show that phycobiliprotein content and composition respond sensitively to changes in light quality and intensity (Gallina *et al.* 2024).

Phycobilisomes are not restricted to cyanobacteria but also constitute major peripheral antenna systems in red algae and glaucophytes. In red algae, the antenna is generally enriched in phycoerythrin, with phycocyanin occupying intermediate positions and allophycocyanin forming the core that transfers excitation toward the photosystems. Changes in the PE ratio, bilin composition, phycobilisome size, and coupling to the photosystems enable acclimation to variations in irradiance and spectral quality. Under low-light or spectrally filtered conditions, an increased contribution of phycoerythrin enhances the capture of green and blue-green photons. Red algae lack the cyanobacterial Orange Carotenoid Protein-dependent quenching system and instead employ lineage-dependent photoprotective mechanisms, including phycobilisome mobility and state transitions in some mesophilic species, partial energetic decoupling of the antenna, and Δ pH-dependent excitation quenching associated predominantly with the PSII core antenna.

Glaucophytes retain cyanobacteria-like phycobilisomes within their plastids, or cyanelles, with phycocyanin and allophycocyanin serving as the principal antenna pigments in well-studied species such as *Cyanophora paradoxa*. These antennae transfer excitation to both PSII and PSI. In *C. paradoxa*, redox-dependent state transitions and redistribution of phycobilisome excitation between the two photosystems represent major short-term mechanisms for limiting excitation imbalance, whereas long-term acclimation to different light spectra involves adjustment of the phycobilisome-to-photosystem ratio and modification of energy-transfer pathways. Thus, although phycobilisome-based light harvesting is conserved among cyanobacteria, red algae, and glaucophytes, the mechanisms used to regulate antenna excitation and prevent photodamage are strongly lineage-specific (Kaňka *et al.* 2014, Misumi and Sonoike 2017, Green 2019, Fang *et al.* 2022, Voerman *et al.* 2022, Ueno and Akimoto 2024).

A well-characterized mechanism is complementary chromatic acclimation (CCA), in which the bilin-bearing phytochrome-class sensor kinase RcaE detects light quality by reversible photoswitching between green-absorbing Pg and red-absorbing Pr states. The signal is then transmitted through a multistep RcaE/RcaF/RcaC phosphorelay, ultimately regulating transcription through the DNA-binding response regulator RcaC. Under red light, phosphorylated RcaC activates transcription of the inducible *cpcB2A2* operon through L-box elements, thereby promoting PC synthesis. Under green light, dephosphorylated RcaC releases repression of *cpeC*, which initiates CpeR-mediated induction of *cpeBA* and leads to PE apoprotein accumulation (Ariyanti *et al.* 2021, Espinoza-Corral *et al.* 2024). This reciprocal regulatory pattern, in which red light favors PC accumulation

whereas green light favors PE accumulation, was initially demonstrated in *Fremyella diplosiphon* and later confirmed in *Calothrix* and other filamentous cyanobacteria. These findings support CCA as a conserved strategy for spectral acclimation in cyanobacteria (Terauchi *et al.* 2004, Prabha *et al.* 2023). These responses include adjustments to photon limitation and spectral shifts that affect excitation distribution within the antenna system.

Under low irradiance, enhanced PE synthesis in *Rhodomonas salina* is viewed as a mechanism that maintains photosynthetic efficiency under energy limitation, when higher light collection efficiency compensates for reduced photon flux (Xie *et al.* 2021). In addition, PE has been associated with antioxidant effects that may increase tolerance to oxidative and temperature stress (Hatha and Sumayya 2023). Notably, thermophilic cyanobacteria from hot springs produce substantially more chlorophyll *a*, carotenoids, and phycocyanin than mesophilic strains (Jaiswal *et al.* 2024). This example indicates that PE-containing complexes can retain functional and applied relevance under regimes where temperature becomes a defining element of the stress profile.

In scenarios accompanied by excess excitation energy, the role of PBPs shifts from compensating energy deficit to limiting photodestructive consequences of overload. High irradiance, nutrient imbalance, and the presence of pollutants increase the risk of photoinhibition and reactive oxygen species formation, especially in organisms with a large antenna. Under these conditions, adaptive value lies in controlling energy transfer at the antenna level, where part of the excitation is redistributed or dissipated before reaching reaction centers, thereby reducing the probability of initiating photooxidative cascades (Grettenberger *et al.* 2024, van Stokkum *et al.* 2025). Further evidence for functional plasticity of the phycobiliprotein system is provided by reports of APC variants that absorb in a more far-red region, expanding possible trajectories of energy redistribution in specialized cyanobacteria adapted to shaded and spectrally filtered habitats (Soulier and Bryant 2023). These findings indicate that PBPs function as a regulated antenna layer capable of modulating the efficiency and direction of excitation transfer in response to changing environmental constraints.

PBPs are also integrated into stress responses through redox-associated effects and the metabolic economy of the cell. PC and PE exhibit antioxidant activity and may contribute to maintaining cellular redox homeostasis under oxidative pressure. In experiments involving heavy metals, xenobiotics, microplastics, and elevated temperature, changes in PBP content often correlate with tolerance, yet the direction of the effect depends on exposure intensity and duration. A biphasic pattern in which PBPs are maintained or increase at early stress stages, but undergo selective degradation under prolonged or strong exposure, is consistent with a trade-off strategy. Within this logic, PBPs initially support energy transfer and redox buffering, and are later mobilized as a nitrogen and carbon source during a transition toward resource reallocation (Kesheri *et al.* 2021). Thus, shifts in the phycobiliprotein pool reflect not only photobiophysical tuning, but also a regulated

balance between preserving photosynthetic function and sustaining metabolic resilience.

The structural implementation of PBP functions is linked to their assembly into large light-harvesting complexes, phycobilisomes, which couple the phycobiliprotein antenna to photosystems. These assemblies are characterized by large molecular size and by high architectural variability depending on environmental conditions (Zheng *et al.* 2021). Canonically, APC forms central antenna components, whereas PC and PE occupy more peripheral positions. Linker proteins define complex geometry and strongly influence energy transfer efficiency (Espinoza-Corral *et al.* 2024). Changes in light intensity, spectrum, and nutrient availability drive remodeling of antenna size, PBP composition, and expression of linker components, which alters energy partitioning between PSI and PSII (Kirilovsky and Büchel 2019). Experiments with mutations in linker components such as ApcE show that disrupting antenna attachment to the thylakoid membrane leads to redistribution of excitation energy, which supports a regulatory role of phycobiliprotein organization in controlling photosynthetic fluxes (Zlenko *et al.* 2019).

Phycobiliprotein gene expression in cyanobacteria is subject to global nitrogen-responsive transcriptional control centred on NtcA, a CAP/CRP-family regulator that binds the palindromic consensus sequence TGTA-(N₈)-TACA within promoters of nitrogen-assimilation genes, including *glnA*, and co-activates operons such as the nitrate assimilation *nir* cluster in concert with NtcB (Jiang *et al.* 2000). Under conditions of nitrogen sufficiency, NtcA represses the GS-inactivating factor genes *gifA* and *gifB*, maintaining glutamine synthetase activity and nitrogen flux; as nitrogen status deteriorates, NtcA-dependent transcriptional reprogramming contributes to the downregulation of biosynthetic capacity, coordinated with the NblS–NblR cascade that drives phycobilisome degradation (García-Domínguez *et al.* 2000). Although NtcA recognition motifs have been identified upstream of the *cpcBA*, *cpeBA*, and *apcAB* operons, direct evidence demonstrating NtcA binding to these promoters and activation of their transcription remains limited. Similarly, while 2-oxoglutarate (2-OG) is widely recognized as a key signal of carbon–nitrogen imbalance and an activator of NtcA, its role as the immediate trigger of phycobiliprotein gene expression has not been experimentally confirmed in the studies examined here. Therefore, the proposed regulatory relationship warrants further experimental validation (Luque *et al.* 2001). The proteolytic arm of this regulatory network operates through the NblS–NblR–NblA cascade. NblS, a membrane-associated PAS-domain histidine kinase, integrates high-irradiance and macronutrient-deprivation signals and transmits bleaching cues through interactions with response regulators RpaB and SrrA; *nblS* mutants fail to regulate *hliA*, *psbA*, and *cpcBA* expression and cannot degrade phycobilisomes under nutrient stress (van Waasbergen *et al.* 2002). NblR promotes *nblA* transcription and is essential for phycobilisome degradation. Its regulatory activity, however, appears to be phosphorylation-independent, as replacement of Asp57 with Ala preserves

normal *nblA* induction and bleaching. Under nitrogen stress, full induction of *nblA* requires coordinated regulation by NblR and NtcA, thereby linking phycobilisome degradation to the NtcA-centered network that senses cellular nitrogen status (Kato *et al.* 2008). NblA1/NblA2 then act as an obligate heterodimeric proteolytic adaptor, binding phycocyanin β -subunits within peripheral rods through conserved N- and C-terminal motifs while simultaneously recruiting ClpC to deliver rod substrates for degradation, producing temporally ordered antenna dismantling in which PC-containing rods are selectively eliminated before NblA-dependent APC core degradation proceeds. Cyanobacteria deficient in *nblA1* and *nblA2* retain intact phycobilisomes during nitrogen starvation, enter a nondividing dormant state, and exhibit pronounced susceptibility to photoinhibition. This phenotype underscores the functional necessity of regulated antenna downsizing under conditions of sustained excitation pressure (Karradt *et al.* 2008).

Across oxygenic phototrophs, similar nonphotochemical quenching phenotypes can arise from fundamentally different antenna-control modules, and this distinction is essential when cyanobacterial responses are compared with those of plants and green algae.

In plants, the dominant energy-dependent NPQ component is driven by lumen acidification: Δ pH formation under excess light conditions has been experimentally linked to activation of NPQ (Steen *et al.* 2020), while mechanistic analysis shows that this process depends on pH-sensitive antenna regulation (Nicol and Croce 2021). Low pH triggers a PsbS conformational response (Krishnan-Schmieden *et al.* 2021), consistent with structural determinants resolved for PsbS (Fan *et al.* 2015), and promotes PSII–antenna interaction changes associated with energy dissipation (Correa-Galvis *et al.* 2016), while xanthophyll-cycle pigments tune the amplitude and kinetics of quenching under fluctuating light (Steen *et al.* 2020).

In green algae, lumen acidification also gates NPQ (Dinc *et al.* 2016), but pH sensing and dissipation are mechanistically integrated into LHCSR proteins (Ballottari *et al.* 2016). LHCSR1/LHCSR3 impose pH-dependent dissipation *in vivo* (Dinc *et al.* 2016, Troiano *et al.* 2021), and zeaxanthin can provide an additional, partly independent quenching route. Thus, convergent photoprotective outcomes arise from fundamentally different antenna architectures and molecular control points across oxygenic phototrophs.

Within this lineage-specific framework, cyanobacterial NPQ relies on the Orange Carotenoid Protein (OCP), which functionally links the phycobiliprotein antenna to energy-dependent dissipation. Under blue-green light, OCP undergoes a structurally defined photoactivation sequence in which the 3'-hydroxyechinenone (3hECN) carotenoid, initially spanning both the C-terminal domain (CTD) and N-terminal domain (NTD) and stabilized by hydrogen bonds to Tyr201 and Trp288, translocates 12 Å into the NTD following photon-driven rupture of these CTD contacts, passing through the intermediate P₃ state before full domain separation on the millisecond timescale

yields the quenching-competent red form (OCP^r) (Andreeva *et al.* 2022). OCP^r binds to the phycobilisome and promotes dissipation of excess excitation energy; however, OCP–phycobilisome association alone is not sufficient to establish the quenched state, indicating that additional conformational rearrangements are required for efficient energy dissipation (Kirilovsky and Kerfeld 2012, Lou *et al.* 2020, Domínguez-Martín *et al.* 2022). Ultrafast spectroscopic and high-resolution structural analyses further show that phycobilisome binding induces rearrangement of the OCP-associated ketocarotenoid and facilitates excitation-energy transfer from phycocyanobilins to the carotenoid, providing a molecular basis for OCP-mediated quenching (Liguori *et al.* 2024, Sauer *et al.* 2024). The contribution of this mechanism increases under high irradiance and oxidative stress, when the probability of photoinhibition and ROS load rises, and experimental studies show that higher OCP levels enhance antenna-dependent photoprotection (Tiwari *et al.* 2025). Increased sensitivity of OCP-deficient strains to photoinhibition and accelerated phycobiliprotein degradation further support the view that the OCP module is a critical element of antenna stabilization under stress exposure (Niziński *et al.* 2025).

Quantitative *in vivo* studies show that OCP-dependent quenching of phycobilisomes is not limited to a decrease in fluorescence, but leads to a measurable redistribution of excitation energy reaching the photosystems. OCP activation reduces the effective absorption cross-section of PSII and modifies Q_A redox kinetics, indicating changes in excitation pressure within the photosynthetic apparatus. Importantly, at physiological OCP levels, this regulation involves only part of the antenna, with approximately half of the phycobilisomes associated with PSII participating, which sets an upper limit to antenna-based energy dissipation (Protasova 2021).

Changes in OCP abundance demonstrate that cyanobacterial photoprotection is constrained by a balance between energy dissipation and photosynthetic efficiency. Elevated OCP levels strengthen thermal dissipation, suppress singlet oxygen formation, and facilitate PSII repair under high-light stress (Sedoud *et al.* 2014, Takahashi *et al.* 2019). In contrast, excessive quenching decreases excitation transfer from phycobilisomes to PSII and consequently limits photosynthetic performance (Ogawa *et al.* 2025). On short timescales following illumination, OCP-dependent NPQ develops in parallel with adjustments in linear and cyclic electron flow, accompanied by measurable shifts in P₇₀₀ redox state, directly linking antenna regulation to electron transport control (Elanskaya *et al.* 2024). The cellular pool of OCP available for photoprotection is subject to multilayered regulatory control. In *Synechocystis* sp. PCC 6803, *ocp* expression is regulated by RpaB, an OmpR-family response regulator that functions within the NblS/Hik33–RpaB signaling pathway and controls numerous genes involved in stress acclimation, photoprotection, and pigment metabolism. Under low-light conditions, phosphorylated RpaB binds HLR1 promoter motifs and represses target gene expression, whereas high-light

exposure promotes RpaB dephosphorylation and rapid transcriptional derepression. This regulatory mechanism is supported by ChIP analyses and by the rapid induction of well-characterized RpaB target genes, including *hliA* and *rpoD3* (Bilger *et al.* 2024). Direct evidence linking RpaB occupancy to the *ocp* promoter has not been reported, and post-transcriptional repression through the sRNA ApcZ provides an independent layer of control over OCP accumulation under non-stress conditions. Regardless, the primary mechanism by which high light scales OCP activity is post-translational: blue-green photon absorption drives structural conversion of OCP^o to OCP^r, making the photoprotective response reactive in character and governed by direct photon sensing at the protein level rather than by transcriptional pre-amplification of the OCP pool (Salvadori *et al.* 2024). The best-supported mechanisms of phycobilisome regulation include complementary chromatic acclimation, OCP-dependent excitation quenching, state transitions, and NblA-mediated antenna degradation. However, similar decreases in phycobiliprotein content may indicate regulated antenna downsizing, nutrient remobilization, or irreversible damage, and these alternatives cannot be distinguished from pigment measurements alone. Functional phycobilisome uncoupling and the distinct strategies of red algae, glaucophytes, and phycobilisome-lacking cyanobacteria further demonstrate that no single photoprotective model applies to all lineages. Future studies should combine spectroscopy, antenna–photosystem coupling measurements, photosystem kinetics, and oxidative-stress indicators.

Functional uncoupling of phycobilisomes from the photosystems represents an additional photoprotective mechanism in cyanobacteria. Under high-light stress, reduced energetic coupling limits excitation transfer from the antenna to the reaction centers, thereby decreasing excitation pressure and photodamage, although at the cost of reduced light-harvesting efficiency. This response is distinct from OCP-mediated thermal quenching, even though both mechanisms reduce excitation delivery to the photosystems (Tamary *et al.* 2012).

Not all cyanobacteria possess phycobilisomes. *Prochlorococcus* instead uses chlorophyll *a*₂/*b*₂-binding Pcb antenna proteins and therefore lacks phycobilisome-dependent OCP quenching. In this genus, zeaxanthin accumulation contributes to photoprotection through antioxidant activity and possibly through a screening function associated with pigment located outside the photosynthetic thylakoid complexes. Diel increases in the zeaxanthin-to-chlorophyll ratio support its protective role under high daytime irradiance, although the precise cellular localization and mechanism require further confirmation (Claustre *et al.* 2002).

Scytonemin as an extracellular UV protection mechanism in cyanobacteria

Scytonemin is a specialized photoprotective pigment of cyanobacteria and fundamentally differs from intracellular systems that regulate light harvesting or excitation

quenching within the photosynthetic apparatus. It is a yellow-brown lipophilic indole pigment deposited not in thylakoids but in the extracellular polysaccharide sheath of some cyanobacteria, especially in lineages inhabiting environments with high and prolonged solar radiation (Mazur and Ślesak 2025). Due to this localization, scytonemin performs a preventive function. It intercepts ultraviolet radiation before it reaches photosynthetic complexes and genomic DNA, thereby reducing the incoming dose of damaging light already at the cell boundary. Phylogenetically, scytonemin occurs across diverse cyanobacterial lineages and has long been recognized as a widespread protective pigment associated with exposure to high ultraviolet radiation (Brenowitz and Castenholz 1997).

The high effectiveness of this strategy is linked to the chemical nature of scytonemin and its optical properties. The scytonemin molecule is a homodimer built from two cyclopentyl indole fragments connected by a carbon–carbon bond. The conjugated ring system provides intense absorption from UV-A to the violet-blue region, approximately 325–425 nm. The absorption maximum is observed at around 370 nm *in vivo* and around 384 nm in organic solvents (Gao *et al.* 2021). Scytonemin can absorb up to 90% of incident UV-A, which is associated with reduced UV-induced inhibition of photosynthetic oxygen evolution (Mazur and Ślesak 2025). Although the main contribution is in UV-A, scytonemin can also substantially attenuate shorter-wavelength regions, including UV-B and even UV-C, which broadens its protective range under natural conditions where radiation spectrum and intensity vary over time and space (Edwards *et al.* 2000).

A key advantage of scytonemin is the combination of shielding capacity and exceptional photostability. Long-term irradiation experiments show that scytonemin persists after months of continuous UV-A exposure, with a degradation rate that is several orders of magnitude lower than that of chlorophylls (Assunção *et al.* 2022). This has direct physiological relevance, because prolonged episodes of high radiation often coincide with desiccation or other states in which energy-dependent photoprotection and enzymatic antioxidant systems operate in a limited manner. Under such regimes, a stable extracellular filter continues to reduce the damaging light dose without requiring active metabolism (Fuentes-Tristan *et al.* 2019).

The ecological role of scytonemin is expressed not only through self-shielding, but also through its integration into cellular stress response pathways. Its synthesis is induced by UV-A and can be enhanced under additional physicochemical stressors, including high salinity, elevated temperature, and desiccation, which indicates coordination of this protection with broader regulatory networks of resilience (Rastogi *et al.* 2015, Cepoi 2024). At the molecular level, UV-A induction in *Nostoc punctiforme* is regulated by the two-component system encoded by *Npun_F1277* and *Npun_F1278*, which is located immediately upstream of the scytonemin biosynthetic cluster. The response regulator *Npun_F1278* is essential for scytonemin production, since deletion of *Npun_F1278* completely abolishes its synthesis. Both

Npun_F1277 and *Npun_F1278* are co-transcribed and induced by UV-A, UV-B, and high visible light. A related UV-A-responsive pathway operates through UirS, an atypical membrane-bound CBCR-class photoreceptor that absorbs UV-A, autophosphorylates, and transfers the phosphoryl group to the AraC-family response regulator UirR, which binds DNA and regulates downstream transcriptional programmes; although published literature connects this phosphorelay to scytonemin cluster activation, direct experimental evidence linking UirR occupancy to *scyABCDE* promoters was not recovered in the primary dataset and warrants independent verification (Soule *et al.* 2009). Scytonemin accumulation is enhanced not only by UV radiation but also by oxidative and osmotic stress. Synergistic induction under combined UV-A and photooxidative conditions has been reported in *Chroococcidiopsis* and *Scytonema*, suggesting that scytonemin biosynthesis is regulated by multiple environmental inputs rather than by light signals alone. These observations support the existence of convergent regulatory pathways through which redox-related signals may influence the scytonemin biosynthetic programme in addition to the primary light-responsive two-component system. Although a MarR-family redox-sensing regulator (RosR) has been proposed as a potential mediator of H₂O₂-dependent activation, direct evidence linking RosR or other MarR-family regulators to scytonemin cluster expression in cyanobacteria remains lacking. Consequently, this component of the regulatory model requires further experimental validation (Rastogi *et al.* 2014). Genera that are rich in scytonemin include *Nostoc*, *Scytonema*, *Calothrix*, *Lyngbya*, *Anabaena*, *Nodularia*, *Chroococcidiopsis*, *Gloeocapsa*, and *Gloeocapsopsis* (Casero *et al.* 2024).

For correct interpretation, it is important to distinguish scytonemin from other UV-protective metabolites of cyanobacteria, primarily mycosporine-like amino acids. These compounds are water-soluble, colorless, and efficiently absorb in the UV-A and UV-B regions, dissipating energy as heat without generating reactive oxygen species, and they may contribute not only to photoprotection, but also to antioxidant defense and osmoregulation (Peng *et al.* 2023). Scytonemin, in contrast, is lipophilic, localized extracellularly, and is particularly effective as an entry barrier, which becomes critical under dehydration and metabolic pause when intracellular protective mechanisms function in a limited manner. Therefore, these compound classes should be viewed not as interchangeable, but as potentially complementary components of a multilevel defense distributed between the extracellular sheath and intracellular compartments.

Beyond its ecological function, scytonemin attracts attention due to biological activity and potential applied value. Alongside its role as a physical UV filter, it exhibits antioxidant activity and can limit UV-induced DNA damage (Pathak *et al.* 2020). Effects in biomedical models have also been reported, including anti-inflammatory and antiproliferative activity, which motivates interest in dermatological and pharmaceutical applications (Rastogi *et al.* 2015). These observations support the idea that

scytonemin is not only an ecological adaptive trait, but also a potential source of molecules with high stability and functional value (Sandybayeva *et al.* 2022). From an engineering perspective, however, scytonemin remains comparatively underexploited. Although elements of its biosynthetic pathway are becoming clearer, broad implementation of heterologous expression and metabolic engineering is still limited, and scaling production is constrained by difficulties in cultivation and pigment extraction (Ferreira and Garcia-Pichel 2016, Rana *et al.* 2026).

Scytonemin extends the acclimatory range by lowering the UV dose before intracellular photodamage develops. Because its shielding effect can persist during desiccation or metabolic dormancy, it complements energy-dependent mechanisms such as NPQ and OCP. However, its quantitative contribution relative to other UV-protective systems under combined stress remains insufficiently resolved. It is an exclusively preventive acclimatory mechanism: it reduces the UV dose reaching intracellular targets before any active photoprotective circuit is engaged, and unlike energy-dependent systems (NPQ, OCP, VDE), it does not require metabolic activity to maintain its protective effect (Pathak *et al.* 2020). This suggests that its protective contribution remains relevant under desiccation, metabolic dormancy, and extreme cold, when energy-dependent photoprotective systems are likely to operate with reduced efficiency rather than as fully active acclimatory mechanisms. Under changing stress conditions in which UV exposure coincides with these metabolic constraints, scytonemin effectively raises the threshold at which the intracellular pigment systems are forced from acclimatory into damage mode, functioning as a passive buffer that extends the acclimatory window for all downstream protection mechanisms. This architecture is particularly valuable when radiation fluctuations coincide with dehydration, salt stress, or temperature extremes, and when active photoprotective mechanisms become less reliable. Therefore, scytonemin can be considered both a marker and a functional resilience module that complements intracellular pigment remodeling and expands the range of conditions under which phototrophs maintain structural integrity and photosynthetic performance.

Evolutionary principles of pigment system remodeling under changing environmental conditions

Natural selection has repeatedly remodeled pigment systems in algae and cyanobacteria, because pigments sit at the interface between energy capture and the constraints of oxygenic photosynthesis. Any increase in absorbed energy raises the risk of photodamage and redox imbalance and therefore makes pigment composition and organization a direct target of selection. In natural habitats, this selection operates under the combined influence of irradiance and photoperiod, nutrient availability, temperature, pH, and salinity, and toxicants, including heavy metals, pesticides, and emerging contaminants. Under such conditions, oxidative stress and disruption of redox homeostasis

become critical constraints for photosystem stability and cellular resilience (Nowicka 2022). Context dependence is especially evident in that superficially similar pigment changes may reflect different biological causes. Under moderate stress pressure, remodeling of the pigment profile often performs a compensatory function and maintains photosynthetic apparatus performance (Foo *et al.* 2020). However, once protective thresholds are exceeded, a different trajectory is observed in which pigment pools decrease due to biosynthetic constraints and increasing damage to antennae by uncontrolled ROS (Anand *et al.* 2021). Therefore, interpretation of pigment shifts along natural gradients requires consideration not only of response direction, but also of load level, exposure duration, and the combination of factors that determine whether the system remains in an acclimation mode or transitions to a degradation mode.

The light environment is the most predictable and persistent source of selective gradients. Its spatial structure is shaped by depth, turbidity, and eutrophication, whereas its temporal structure reflects seasonality, diel cycles, and rapid irradiance fluctuations (Litchman 2000, Tsakalakis *et al.* 2022). Along these gradients, differences in chlorophyll ratios and in accessory pigment composition have evolved, because changes in light spectrum and intensity directly alter the balance between the need for light harvesting and the need to limit excitation pressure (Kübler and Raven 1996). More recent work further shows that variability in the ocean light spectrum can differentially favor phytoplankton with flexible pigment content and chromatic acclimation capacity (Mattei *et al.* 2026).

Subsequent field and laboratory studies expanded this framework by incorporating ultraviolet exposure, life-cycle stage dependence, and stressor interactions. Notably, ultraviolet-driven pigment responses have been documented in natural seasonal settings (Ha *et al.* 2015), whereas combined exposure regimes, including interactions with light–dark cycling, can modify not only the magnitude of the response but also its direction (Aguilera *et al.* 2008).

Photoprotection across these gradients is supported not only by carotenoid mechanisms. Mycosporine-like amino acids and other secondary metabolites, including phenolic compounds, show pronounced sensitivity to ultraviolet regime and seasonality (Dunlap and Shick 1998, Korbbe *et al.* 2005). In some cases, such compounds may also extend protective effects beyond the cell surface through release into the surrounding environment, creating an additional layer of ecological resilience at the community level (Swanson and Druehl 2002). This indicates that selection acts not on individual molecules, but on integrated strategies in which different pigment and metabolite classes functionally complement each other depending on the nature of light and chemical pressure.

Situations in which stress regimes recur consistently and become background conditions for stable differentiation among populations and strains merit special attention. However, the mechanistic links between short-term phenotypic acclimation and long-term genotypic

adaptation remain only partly resolved, particularly for traits related to photosynthesis, growth, and stress tolerance (Aardema *et al.* 2024, Liang *et al.* 2025). This gap is especially apparent for protective pigment accumulation. In *Haematococcus lacustris*, nitrogen starvation combined with high irradiance is a well-established trigger that culminates in the formation of astaxanthin-enriched cysts. However, early protective stages at the onset of stress exposure, before pronounced carotenoid accumulation and morphological transitions, remain less clearly defined than the late phenotype stage (Li *et al.* 2022). Natural gradients indicate that pigment system remodeling is not a universal response following a single regulatory template, but a product of selection acting on functional trade-offs among light harvesting, excitation limitation, redox stabilization, and maintenance of structural integrity of the photosynthetic apparatus. Recurrent general patterns, such as strengthening photoprotective programs under moderate load and reduction of pigment pools when resilience thresholds are crossed, provide a basis for comparing stress scenarios and for explaining why the same pigment markers can indicate either adaptation or damage depending on context (Maltsev *et al.* 2021, Rampengan *et al.* 2025).

Natural models of long-term multifactor selection in extremophilic phototrophic microorganisms

Extremophilic phototrophs provide a window into how pigment systems are shaped under long-term and recurrent exposure to multiple factors acting simultaneously. In contrast to short-term acclimation, such conditions impose stable selection regimes in which value lies not in a single pigment, but in coordinated operation of light harvesting, photoprotection, and redox control across repeated cycles of damage and recovery. Therefore, extreme habitats serve as natural analogs of multifactor stress in which it is possible to trace which elements of the pigment apparatus become prioritized, and which recede while population viability is maintained (Yuan *et al.* 2024).

Desert biological soil crusts illustrate one of the most stringent scenarios of combined pressure. Cyanobacteria in these systems simultaneously experience high UV exposure, chronic desiccation with episodic rehydration, sharp diel and seasonal temperature fluctuations, and often elevated salinity. Resilience is typically provided by a suite of traits that includes extracellular UV-shielding compounds, antioxidant defenses involving carotenoids and enzymatic systems, and extracellular polymeric substances that increase water retention and contribute to salt tolerance (Xu *et al.* 2021). An additional requirement is the ability to enter a state of metabolic dormancy during desiccation and rapidly restore activity after wetting, a feature that has been extensively documented in cyanobacteria inhabiting arid environments (Williams 2006). More recent studies further show that repeated cycles of desiccation and rehydration impose constraints on photosynthetic recovery and require stabilization of pigment systems and photoprotective mechanisms (Romero *et al.* 2020, Ye *et al.* 2021).

The ecological role of biocrusts further links pigment phenotypes to measurable fitness outcomes. Cyanobacteria act as pioneer colonizers, stabilize the substrate through filamentous growth and EPS, facilitate succession, and contribute to nitrogen and carbon cycling (Sepehr *et al.* 2019, Fattahi *et al.* 2020). Their dominance in arid landscapes indicates that the combination of sheath-based and pigment-based strategies, including UV shielding, has direct functional value under extreme insolation and water deficit. At the same time, biocrust communities show specialization. Genera such as *Microcoleus* and *Coleofasciculus* primarily contribute to structural stabilization of the substrate, whereas *Nostoc* and *Scytonema* are associated with nitrogen fixation and enhanced UV tolerance (Zhang *et al.* 2021, Etesami 2025). This pattern emphasizes that pigment adaptations are selected in a community context, where complementary strategies collectively determine ecosystem resilience.

At the level of pigment traits, arid systems show a recurrent evolutionary pattern consistent with the threshold logic discussed above. When the combination of UV exposure and dehydration makes photosynthesis especially risky, selection may favor reduced chlorophyll *a* together with stronger passive screens and protective molecules. In cyanobacteria from biocrusts, drought is often associated with lower chlorophyll *a* and higher scytonemin. In *Scytonema*, pronounced induction of scytonemin has been reported under combined drought and UV exposure (Pathak *et al.* 2022). Under moderate load, compensatory pigment remodeling can occur, whereas extreme or prolonged exposure shifts the system toward antenna limitation and reduction of pigment pools to decrease uncontrolled ROS formation. Similar threshold behavior is also observed in experiments with toxicants. Low doses sometimes maintain or increase pigment levels *via* compensatory or hormetic responses, whereas higher doses reduce pigments due to damage and biosynthetic constraints (Ismaiel and Piercey-Normore 2024, Li *et al.* 2025).

Temperature extremes impose a different type of multifactor pressure because they change the relationship between energy input and energy use and thereby alter the probability of redox imbalance. At low temperatures, photochemical excitation may continue while downstream energy use and repair become constrained, increasing the risk of overexcitation and photooxidative damage, especially under photochilling conditions (Míguez *et al.* 2017, Ermilova 2020). Comparative studies report that cold stress is often associated with activation of photoprotective cycles and changes in pigment ratios, including xanthophyll-cycle indices (Míguez *et al.* 2017). In cyanobacteria, Hik33, a histidine kinase carrying transmembrane, HAMP, and PAS domains that detects low-temperature-induced rigidification of the lipid bilayer, initiates cold perception at the membrane – a mechanism confirmed by FTIR analysis and by the loss of cold-gene induction in desaturase-deficient backgrounds where membrane fluidity cannot change appropriately (Los and Leusenko 2025). Upon activation, Hik33 autophosphorylates and signals through

the response regulator Rre26, controlling over 60% of cold-inducible genes including the fatty acid desaturases *desB* and *desD*, the high-light-inducible proteins *hliA* and *hliB*, and genes governing photosystem stoichiometry and antenna composition (Mikami and Murata 2003). Sigma factor SigD (*rpoD3*) operates in parallel: under stress conditions that accompany cold exposure, including $^1\text{O}_2$ accumulation arising from impaired PSII repair and slowed electron transport, *sigD* is induced as part of the $^1\text{O}_2$ -responsive transcriptome, and SigD-dependent promoter selectivity reshapes transcription of photosynthesis-related and pigment-system genes, a regulatory axis that intersects antagonistically with SigB and expands under oxidative and high-light inputs (Yuan *et al.* 2026). Direct transcriptional induction of *sigD* by low temperature has not been conclusively demonstrated. Current evidence suggests that the observed response is more likely mediated through secondary $^1\text{O}_2$ -dependent signaling pathways than through direct cold perception. However, when cold stress exceeds the capacity of photoprotective acclimation, substantial pigment loss may occur. For example, freezing treatment in *Neoporphyra haitanensis* resulted in reduced F_v/F_m values accompanied by significant declines in chlorophyll *a* and phycobiliprotein content, indicating that the effects of low temperature on pigment systems are strongly dependent on both stress severity and taxonomic background (Zhu *et al.* 2022). Some taxa enhance antioxidant capacity under cold exposure (Ernilova 2020), whereas others suppress photosynthetic investment to limit photooxidative risk (Zhu *et al.* 2022). These contrasting responses reflect taxon- and regime-specific limits of photoprotection, repair, and redox homeostasis rather than a universal cold-stress response. In contrast, hypersaline habitats impose another stable selection regime in which osmotic stress is tightly linked to redox signaling and membrane organization and is therefore especially informative for understanding the evolution of carotenoid solutions. *Dunaliella salina* remains a classic model in which the combination of high salinity, high irradiance, and nutrient limitation leads to pronounced β -carotene accumulation (Goswami *et al.* 2022). Importantly, not only synthesis but also the mode of deposition matters. In *Dunaliella*, stress-induced β -carotene is compartmented in β -carotene-rich plastoglobuli/lipid droplets within the plastid system (Davidi *et al.* 2014). This storage mode functionally connects carotenoid accumulation with triacylglycerol metabolism and carbon-sink formation under nitrogen limitation (Bonnetfond *et al.* 2017). Variation among strains is also informative. Carotenogenic and non-carotenogenic lineages can achieve photoprotection *via* different routes, combining β -carotene deposition with different degrees of reliance on xanthophyll mechanisms, which highlights multiple evolutionary trajectories toward a similar functional task (Souza *et al.* 2024). For *D. salina*, links between ROS and induction of β -carotene under high irradiance have been described, which supports the view of carotenoid phenotypes as products of redox-regulated feedback rather than passive consequences of damage (Ye *et al.* 2008).

Metal-enriched and acidic ecosystems add an additional layer of multifactor pressure because they intensify ROS chemistry and affect metal-sensitive elements of the photosynthetic apparatus. Heavy metals, irrespective of their redox activity, cause toxicity predominantly by inducing oxidative stress *via* direct or indirect mechanisms (Nowicka 2022). For pigment systems, this is especially critical because tetrapyrroles and iron–sulfur centers are vulnerable targets, and their oxidative destabilization disrupts electron transport and amplifies redox imbalance *via* positive feedback (Asril *et al.* 2025). Under such conditions, pigments contribute to resilience through antioxidant and photoprotective functions, yet once protective capacity is exceeded, pigment loss becomes likely because of damage and biosynthetic limitation.

Within this set of models, *Galdieria sulphuraria* is particularly valuable because it combines tolerance to extreme acidity and high temperature (Thangaraj *et al.* 2011) with resistance to a range of heavy metals (Kharel *et al.* 2024). These conditions increase pressure on photosynthetic functions dependent on chlorophyll/tetrapyrrole integrity and Fe–S cluster assembly, processes that are widely metal-sensitive and closely linked to photosynthetic stability. Model relevance is also linked to features of photosynthetic-apparatus organization that can support efficiency under low irradiance and, under resource limitation, may be accompanied by a reduced share of photosynthetic components. This phenotype is consistent with a conservative logic: under resource scarcity and high redox load, the system limits photosynthetic investment to keep photooxidation risk within tolerable bounds (Stadnichuk *et al.* 2011).

In natural extreme habitats, long-term multifactorial stress selects not for a high-pigment phenotype, but for a stable pigment-system configuration in which light harvesting is continuously matched to cellular redox capacity and thylakoid membrane state. The most reproducible outcome is a shift in emphasis from antenna maximization to risk control through radiation screening, switchable energy dissipation, antioxidant buffering, and structural stabilization. Therefore, pigment variation across natural gradients should be interpreted as a marker of selected survival strategies that distinguish compensatory remodeling modes from protective limitation modes, rather than as a universal indicator of photosynthesis enhancement or suppression.

Experimental evolution as a causal approach to identify heritable remodeling of pigment systems

Long-term evolution experiments (LTEE) and adaptive laboratory evolution (ALE) provide a methodologically rigorous opportunity to distinguish short-term acclimation of pigment systems from heritable adaptation that is fixed across generations. In natural gradients of combined stress, light regime, nutrient availability, temperature, salinity, and toxicants often change jointly; therefore, observed changes in pigments and photobiophysical parameters often remain correlational and allow multiple interpretations. In ALE and LTEE, selective pressure is

predefined and maintained over long periods, which makes it possible to test causal links between a stress regime, pigment phenotype dynamics, and trait stability after re-adaptation or transfer between conditions (Sun *et al.* 2018, LaPanse *et al.* 2021).

A key advantage of ALE and LTEE is their temporal resolving power. Early phases are dominated by rapid physiological responses, including remodeling of light-harvesting architecture and changes in nonphotochemical quenching (NPQ), comprising the fast-reversing energy-dependent component (q_E ; ΔpH /xanthophyll-dependent), state transitions (q_T ; LHCII phosphorylation *via* STN7/STT7 kinase), photoinhibitory quenching (q_I ; slow-reversing, linked to D1 turnover), and the PsbS-independent zeaxanthin-dependent component (q_Z), whereas at later stages under sustained stress, stable changes in regulation and metabolic coupling can become fixed, appearing as stable pigment ratios and reproducible profiles of individual pigments. For this reason, causal interpretation requires no single time points but comparable time series, replicate lines, and standardized tests of reversibility (Lakeman *et al.* 2009, Zhao and Huang 2021). This logic is summarized in Fig. 2, which shows how a defined stressor

is sequentially translated into proximate constraints on photosynthesis, triggers remodeling of pigment modules, and is then fixed as a heritable phenotype in ALE and LTEE designs.

Empirical LTEE outcomes under high irradiance indicate that stable phenotypes often arise not through simple increases in the total pigment pool, but through optimization of light harvesting relative to photodamage risk. Selection under extremely high irradiance up to $3,000 \mu\text{mol}(\text{photon}) \text{m}^{-2} \text{s}^{-1}$ led to tolerant lines, which indicates a directed nature of long-term light pressure (He *et al.* 2025). In the study by Bonnefond *et al.* (2022), after 160 d of selection under dynamic light, productivity increased by 77% together with an 85% reduction in antenna cross-section and the appearance of 2,700 new alleles, which is consistent with selection for architectural solutions that reduce excitation pressure. Complementarily, Russell *et al.* (2022) showed a 1.9-fold increase in β -carotene production in *Tetraselmis tetrathele* under high light, which corresponds to strengthened photoprotective carotenoid buffering under chronic photooxidative load.

Salt stress highlights the need to distinguish strictly between plasticity and heritable adaptation. Acute salinity

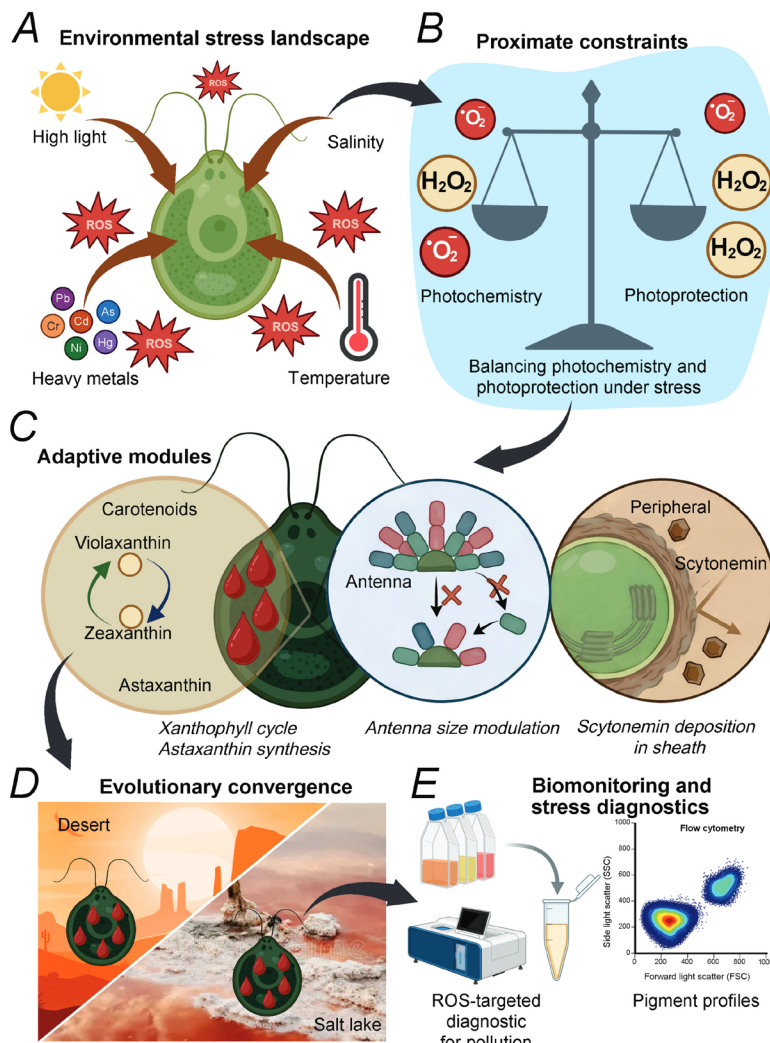


Fig. 2. Causal framework of ALE and LTEE to separate acclimation and heritable adaptation of pigment systems under chronic stress in microalgae. (A) Definition of a selective regime that includes a specific stressor or its combination, for example, high light, salinity, heavy metals, or their combinations. (B) Formation of proximate photobiophysical constraints, including increased excitation pressure, increased probability of reactive oxygen species (ROS) formation, and shifts in redox balance. (C) Short-term acclimation manifested as changes in NPQ, redistribution of excitation energy, and early shifts in pigment ratios. (D) Multigenerational selection in replicate lines with accumulation of heritable changes in regulation and metabolic coupling that affect pigment profile stability. (E) Verification of heritability through re-adaptation, condition transfer, and comparison of ancestral and selected phenotypes, including measurements of pigments, PSII parameters, NPQ, and oxidative stress markers.

shifts trigger immediate osmotic effects and growth inhibition that are typically interpreted as acclimatory responses, whereas heritable adaptation requires long-term selection and explicit separation of acclimation from evolutionary change (Borowitzka 2018, Melero-Jiménez *et al.* 2022). Farkas *et al.* (2023) demonstrated pronounced structural plasticity in green microalgae at 0.1–1 M salinity, including palmelloid-like stages in *Chlamydomonas reinhardtii* CC124, marked changes in cell size in *Chlorella* sp. MACC-360, and salt-associated remodeling of thylakoid organization. These observations are highly informative as early stress responses; however, without a multigenerational selection framework, they should not be interpreted as evidence of heritable pigment adaptation.

Metal stress is particularly informative for causal analysis because it directly perturbs photosynthetic electron transport, enhances ROS generation, and destabilizes pigment homeostasis. Under these conditions, pigment metabolism represents part of redox homeostasis rather than only a damage target (Tripathi and Poluri 2021, Nowicka 2022). At the same time, declines in pigment content do not necessarily scale linearly with PSII functional impairment, because toxic metals can affect different components of the photosynthetic apparatus with unequal sensitivity, and fluorescence parameters may change even when some pigment pools remain comparatively stable (Kumar *et al.* 2014, Dong *et al.* 2022). Nanda *et al.* (2021) reported that under exposure to Cd(II) and Pb(II), carotenoids could be relatively better preserved and, in some settings, could increase at later time points compared with chlorophylls, while pigment ratios changed over time in line with activation of antioxidant programs. Such examples indicate that long-term selection and temporal dynamics are necessary for interpreting pigment phenotypes as elements of stable adaptation.

Current evidence strongly supports scytonemin as a stable extracellular barrier against chronic UV-A exposure, particularly when dehydration or metabolic limitation restricts energy-dependent protection. Its broader regulation by oxidative, osmotic, and temperature stress is plausible, but several proposed transcriptional and redox-control pathways still lack direct experimental validation. We therefore consider scytonemin a long-term protective investment that complements, rather than replaces, rapid intracellular quenching and antioxidant mechanisms. Important unresolved questions include its biosynthetic cost, induction threshold, ecological trade-offs, and quantitative contribution relative to mycosporine-like amino acids under combined environmental stresses.

Convergent functional trajectories of pigment system remodeling based on ALE and LTEE

ALE and LTEE data allow discussion of recurrent trajectories of pigment-system remodeling as convergent solutions that arise under chronic stressor exposure. Convergence here refers to the functional logic of adaptation rather than to a fixed pigment profile across taxa, because under similar selective pressure different

biochemical implementations can achieve equivalent regulation of excitation pressure while maintaining acceptable levels of photooxidative risk.

In a thermally driven ALE of *Picochlorum* sp. BPE23 conducted in a turbidostat under diel light and temperature cycles with stepwise increases of the daytime temperature peak up to 49°C, direct measurements of chlorophylls and carotenoids were not performed, and the pigment component was assessed *via* the specific optical absorption cross-section over 400–700 nm. By the end of selection, this value decreased from 221.9 ± 18.9 to $143.4 \pm 23.3 \text{ m}^2 \text{ kg}^{-1}$, which was interpreted as a regulated reduction of light absorption under thermal stress together with an increased upper temperature limit for growth in the isolated clones (Barten *et al.* 2022).

In an ALE of *Chlorella vulgaris* under stepwise hypersalinity in serial transfer cycles, selected lines E2 and E2.5 were characterized by an approximately 2.2-fold increase in total lipid content while maintaining growth and biomass relative to the control. Total chlorophyll in the best lines did not show a pronounced decline, and oxidative stress markers did not indicate increased lipid peroxidation, which is consistent with resilience forming as a systemic metabolic and redox adaptation rather than as an isolated remodeling of the pigment pool (Varunraj *et al.* 2023).

In a high-light ALE of *Synechocystis* sp. PCC 6803, selection across increasing irradiance levels produced populations able to grow under light intensities that were lethal for the ancestral laboratory type, including population-level resilience at $3,000 \mu\text{mol}(\text{photon}) \text{ m}^{-2} \text{ s}^{-1}$. The genetic architecture of the trait was polygenic, and causal testing of individual reconstructed mutations showed increased growth and carotenoid content at $700 \mu\text{mol}(\text{photon}) \text{ m}^{-2} \text{ s}^{-1}$, while single mutations did not enable survival in $2,000 \mu\text{mol}(\text{photon}) \text{ m}^{-2} \text{ s}^{-1}$, which indicates that combinations of changes are required for the extreme phenotype (Dann *et al.* 2021).

In a salt ALE of *Synechocystis* sp. PCC 6803 in which ten independent lines S1–S10 evolved from a single clone under constant 3% NaCl, lines obtained after 303 d of selection showed improved growth under salt, but divergent metabolomic and transcriptomic response programs. Such divergence under the same selective factor supports the interpretation that convergence can occur at the level of phenotypic function, whereas molecular solutions, including components linked to pigment and redox tuning, remain multiple and line-specific (Hu *et al.* 2020).

Thus, under prolonged selective exposure, different taxa can reach the same functional state in which light-harvesting intensity is adjusted to the limits of stable photochemistry and redox homeostasis. Under thermal regimes, this is expressed as reduced effective light absorption with preserved photosystem performance under elevated temperature load, whereas under extremely high light it is expressed as heritable retuning of photosystem organization and electron transport, often polygenic in nature and linked to the photoprotective contribution of carotenoids. Under salt stress, resilience

is more often fixed as an integrated outcome of resource reallocation and increased antioxidant capacity, with divergence of molecular programs among lines; therefore, pigment metrics are more appropriately treated as part of coordinated systemic adaptation rather than as the sole target of selection.

Pigment metrics in pollution monitoring and control of culture resilience

Pigment profiles of microalgae and cyanobacteria are among the most responsive physiological indicators under heavy metal exposure because they closely track the state of the photosynthetic apparatus, redox balance, and photooxidative load. Toxic metal stress rapidly affects chlorophylls and accessory pigments, as pigment–protein complexes are highly sensitive to disturbances in electron transport and ROS overproduction (Nowicka 2022). Under heavy metal exposure, cells mitigate toxicity and preserve photochemical function through a combination of reduced uptake, active export, intracellular sequestration, and metal chelation, including phytochelatin-based detoxification, together with reinforcement of antioxidant defence (Tripathi and Poluri 2021, Chakravorty *et al.* 2023). These responses are commonly accompanied by changes in antioxidant enzyme activities such as SOD, CAT, and APX, as well as by the mobilization of low-molecular-mass redox buffers, including glutathione, ascorbate, tocopherols, and carotenoids (Tripathi and Poluri 2021, Nowicka 2022). As a result, a measurable physiological response pattern emerges in which changes in the pigment pool represent an integrated consequence of redox-homeostasis disruption and altered energy partitioning rather than a single isolated reaction.

The kinetics of pigment responses make them particularly valuable for early detection of environmental change. Compared with higher organisms and with endpoints that require prolonged effect accumulation, microalgae show rapid shifts in pigment composition as pollutant concentrations change, which supports their use as early indicators of metal load and toxicity levels (Carfagna *et al.* 2013, Wang *et al.* 2019). At the same time, interpretation of pigment metrics requires consideration of dose, duration, and multifactor exposure structure. Inhibitory pollutant concentrations often reduce chlorophyll and carotenoid contents due to biosynthesis suppression, membrane damage, and photosystem destabilization, which is accompanied by increased oxidative damage to lipids and macromolecules (Jamers *et al.* 2013, Japar *et al.* 2021). Nonlinear scenarios can also occur under sublethal loads, where early time points show transient compensatory shifts, including temporary increases in pigment levels and light absorption capacity. Such responses are consistent with hormetic mechanisms and reflect an early phase of the stress program rather than improved ecosystem physiological status; therefore, pigment increases at low doses cannot be treated as a simple equivalent of optimal cultivation conditions (Agathokleous *et al.* 2022, Zhang *et al.* 2024). Under mixed pollutants and concurrent factors such as

changes in light, trophic status, and ionic strength, this dose-dependent ambiguity becomes especially important.

The most reliable diagnostic solutions arise when moving from single indicators to multiparameter panels in which pigments are interpreted together with redox-stress markers and photochemical parameters. Chlorophyll *a*, chlorophyll *b*, and carotenoids remain core metrics because they are linked to carbon fixation efficiency and ATP synthesis and therefore to functional preservation of the photosynthetic apparatus (Chakravorty *et al.* 2023). Diagnostic accuracy increases when these metrics are complemented with lipid peroxidation markers, antioxidant activity parameters, and measures of photosystem efficiency (Chen *et al.* 2022). This approach helps distinguish compensatory acclimation, in which pigment levels remain stable despite elevated redox markers, from conditions in which pigment decline is accompanied by membrane destabilization and reduced photosynthetic activity (Liu *et al.* 2022). Toxicological assessment is therefore more robust when pigment responses are interpreted together with redox and photosynthetic parameters rather than in isolation.

Practical applicability of pigment profiles is evident in both biomonitoring and culture control in complex media. In biosensor configurations, changes in physiological activity, including pigment-linked responses and photochemical parameters, are used as signals of inhibitory factors, and applied systems widely use genera such as *Chlorella*, *Chlamydomonas*, *Arthrospira*, *Nannochloropsis*, and *Scenedesmus* (Zeraatkar *et al.* 2016). Reliability of such approaches depends on calibration to realistic pollutant mixtures and control of factors that can distort the signal, including shading effects and physicochemical properties of nanomaterials whose toxicity strongly depends on size, surface chemistry, aggregation, and exposure duration. In technological systems oriented toward cultivation on wastewater, pigment monitoring can serve as an operational physiological control tool that helps track transitions from a productive state to a pronounced stress mode.

Pigment profiles provide a robust indicator layer because they integrate the consequences of photochemical disruption, redox imbalance, and changes in membrane stability into quantitatively measurable traits. Maximum reliability is achieved when pigment data are interpreted within multiparameter panels aligned with oxidative damage markers and photosynthetic parameters. This approach enables correct handling of nonlinear dose-dependent responses, discrimination of early compensatory shifts from developed damage, and use of pigment data as an early warning and controllable monitoring tool in polluted and wastewater media.

Conclusion

Combined environmental stresses disrupt the balance between absorbed light energy and the capacity of microalgae and cyanobacteria to use or dissipate excitation safely. Pigment remodeling is therefore a central component of stress acclimation rather than merely an indicator of exposure. Chlorophyll systems adjust

antenna size and pigment organization, carotenoids enhance nonphotochemical quenching, antioxidant protection, and membrane stability, while phycobiliproteins regulate spectral light capture, excitation transfer, and resource allocation. Specialized pigments such as scytonemin provide an additional extracellular barrier against chronic ultraviolet radiation. The strongest evidence supports rapid excitation dissipation, antenna resizing, xanthophyll-cycle activity, OCP-mediated quenching, and regulated phycobilisome degradation as major protective mechanisms. However, similar pigment changes may also result from nutrient remobilization or irreversible photodamage. Consequently, total pigment content or a single pigment ratio cannot reliably distinguish acclimation from damage without corresponding measurements of photosystem performance and oxidative injury. The major unresolved challenge is to connect pigment phenotypes with regulatory mechanisms and ecological context. Future studies should use time-resolved, multiparametric designs that combine pigment profiles with PSI and PSII performance, nonphotochemical quenching, ROS, lipid peroxidation, redox status, and transcriptomic or proteomic data under realistic combinations of light, temperature, nutrients, salinity, and pollutants. Such approaches will clarify when pigment remodeling preserves photosynthetic function and when it indicates that cellular protective capacity has been exceeded.

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